

HUMANOID WHOLE-BODY AFFORDANCE DEBT: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

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ABSTRACT

We study humanoid whole-body manipulation. The motivating bet is: Measure and minimize future affordance loss induced by current whole-body posture choices.. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For humanoid whole-body manipulation, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2026a; 2015; 2026b;c; 2025a;b). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.547	0.453
observed_only_wam	hidden_mode_shift	0.168	0.833
augmentation_as_data_wam	nominal	0.588	0.412
augmentation_as_data_wam	hidden_mode_shift	0.260	0.740
scalar_uncertainty_wam	nominal	0.625	0.375
scalar_uncertainty_wam	hidden_mode_shift	0.340	0.660
proposed_branch_mechanism	nominal	0.800	0.200
proposed_branch_mechanism	hidden_mode_shift	0.560	0.440

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Humanoid Whole-Body Affordance Debt argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Whole-body pushing manipulation with contact posture planning of large and heavy object for humanoid robot, 2015. <https://doi.org/10.1109/icra.2015.7139995>.
- Lhm-humanoid: Learning a unified policy for long-horizon humanoid whole-body loco-manipulation in diverse messy environments, 2025a. <http://arxiv.org/abs/2508.16943v2>.
- Humanoid locomotion and manipulation: Current progress and challenges in control, planning, and learning, 2025b. <http://arxiv.org/abs/2501.02116v2>.
- Bifrostumi: Bridging robot-free demonstrations and humanoid whole-body manipulation, 2026a. <http://arxiv.org/abs/2605.03452v1>.
- Humanoid manipulation interface: Humanoid whole-body manipulation from robot-free demonstrations, 2026b. <http://arxiv.org/abs/2602.06643v2>.
- Ultra: Unified multimodal control for autonomous humanoid whole-body loco-manipulation, 2026c. <http://arxiv.org/abs/2603.03279v1>.